Predicting High vs Low Revenue Days in a Cafe

A machine learning approach to predicting daily café revenue using temporal and historical patterns.

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Introduction to Machine Learning
COMP-4002

Problem Statement

Can we predict whether a given day will be a high-revenue or low-revenue day in time, based on temporal features?

This problem is important for business decisions such as staffing, inventory planning, and demand forecasting. Accurately predicting high- and low-revenue days allows café managers to allocate resources more efficiently and reduce operational costs. It also helps improve customer satisfaction by ensuring the right level of service during busy periods

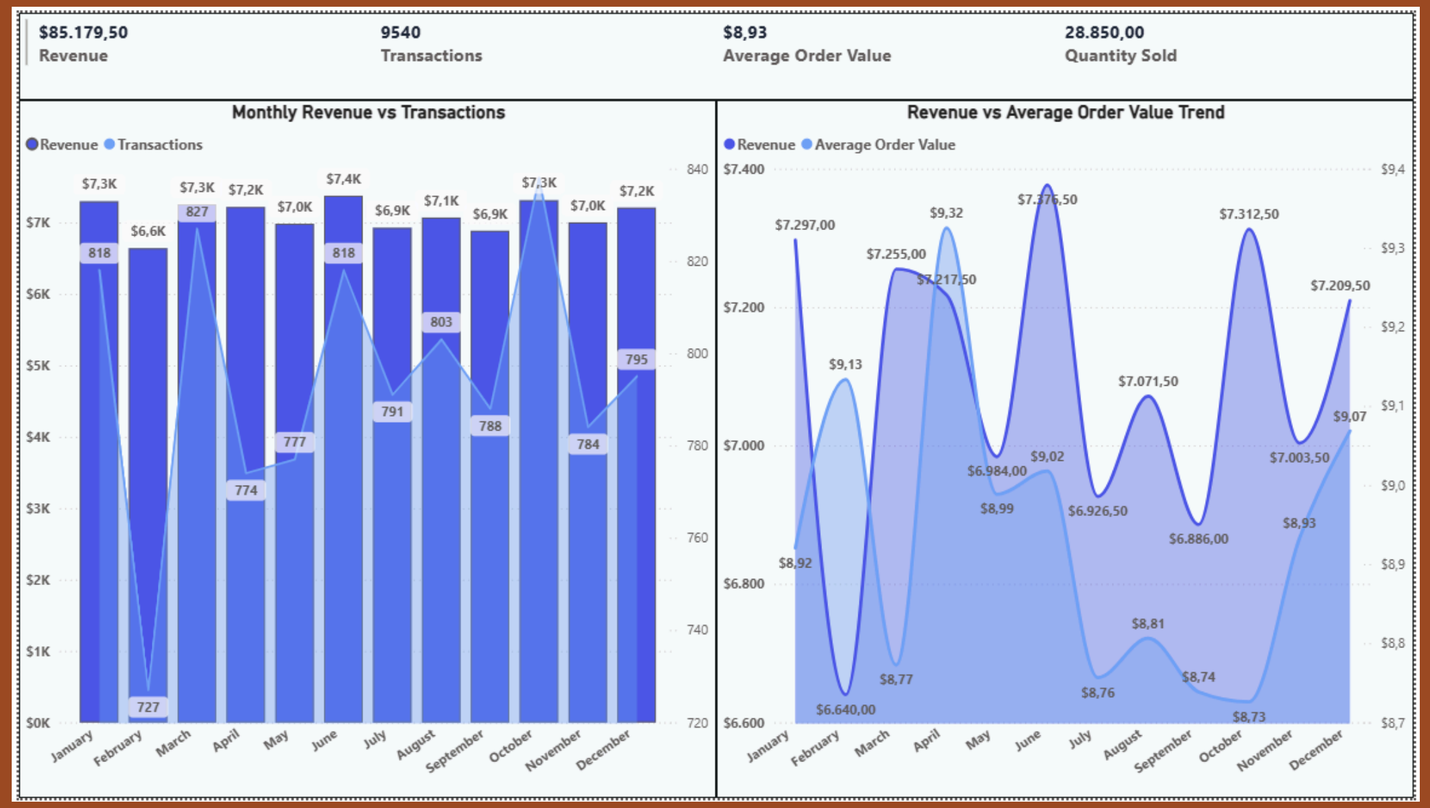
Data Overview

Data Summary

- 9,540 transactions (full year: Jan–Dec 2023)
- 20 features, no missing values
- 8 menu items

Transaction Characteristics

- Quantity: 1–5 items per order
- Price: £1–£5 per item
- Total spend: £1–£25 per transaction
- Avg order value: £8.93
- Avg daily revenue: ~£233



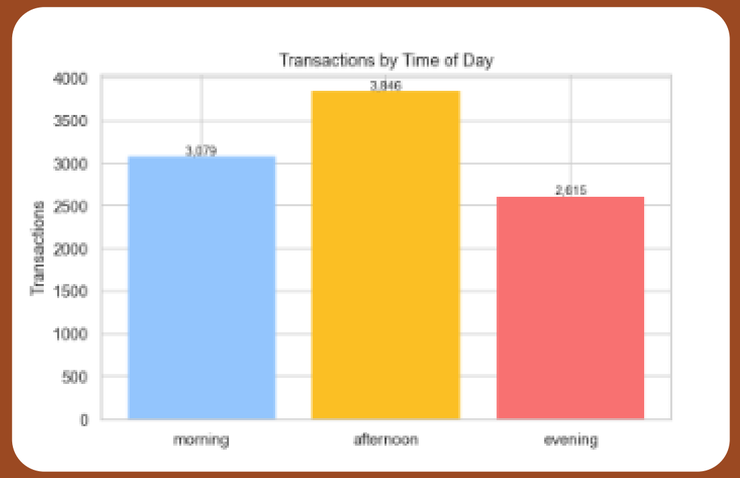
Cleaning of Data

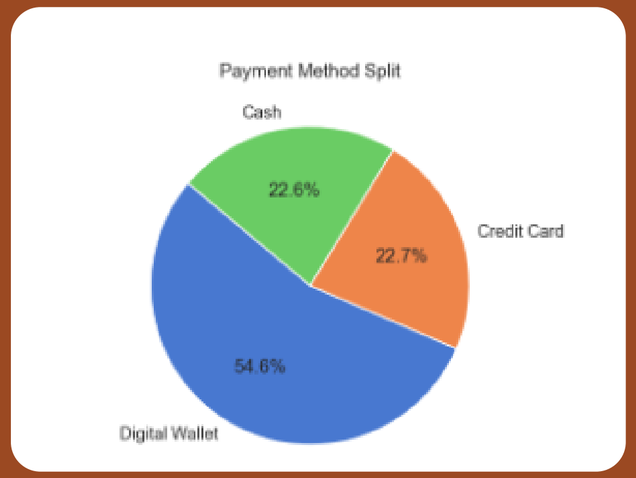
- Converted data types:
  - Numeric → pd.to_numeric(errors="coerce")
  - Dates → pd.to_datetime(errors="coerce")
  - Categorical → category dtype
- Checked for duplicates:
  - No duplicate rows or Transaction IDs
  - All 9,540 records retained
- Handled outliers using IQR-based capping:
  - Applied to Quantity, Price Per Unit, and Total Spent
  - No significant outliers in Quantity & Price (bounded 1–5)
  - Minor extreme values in Total Spent (>£25) were capped

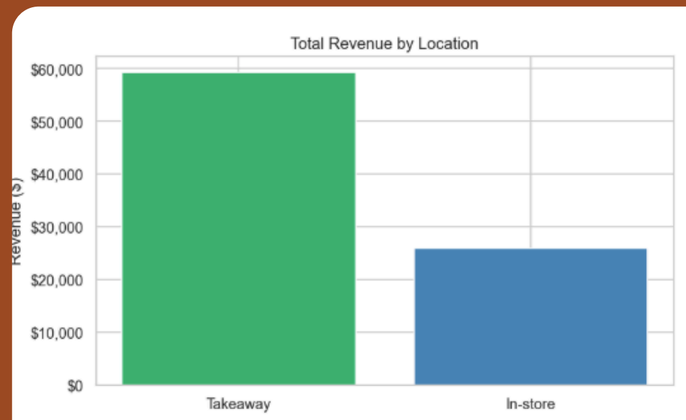
Data Overview

Customer Behaviour

- LocationTakeaway: 70%
- In-store: 30%
- PaymentDigital Wallet: 55%
- Credit Card: 23%
- Cash: 23%
- Time of DayAfternoon: 40% (peak)
- Morning: 32%
- Evening: 27%
- Week SplitWeekdays: 71%
- Weekends: 29%



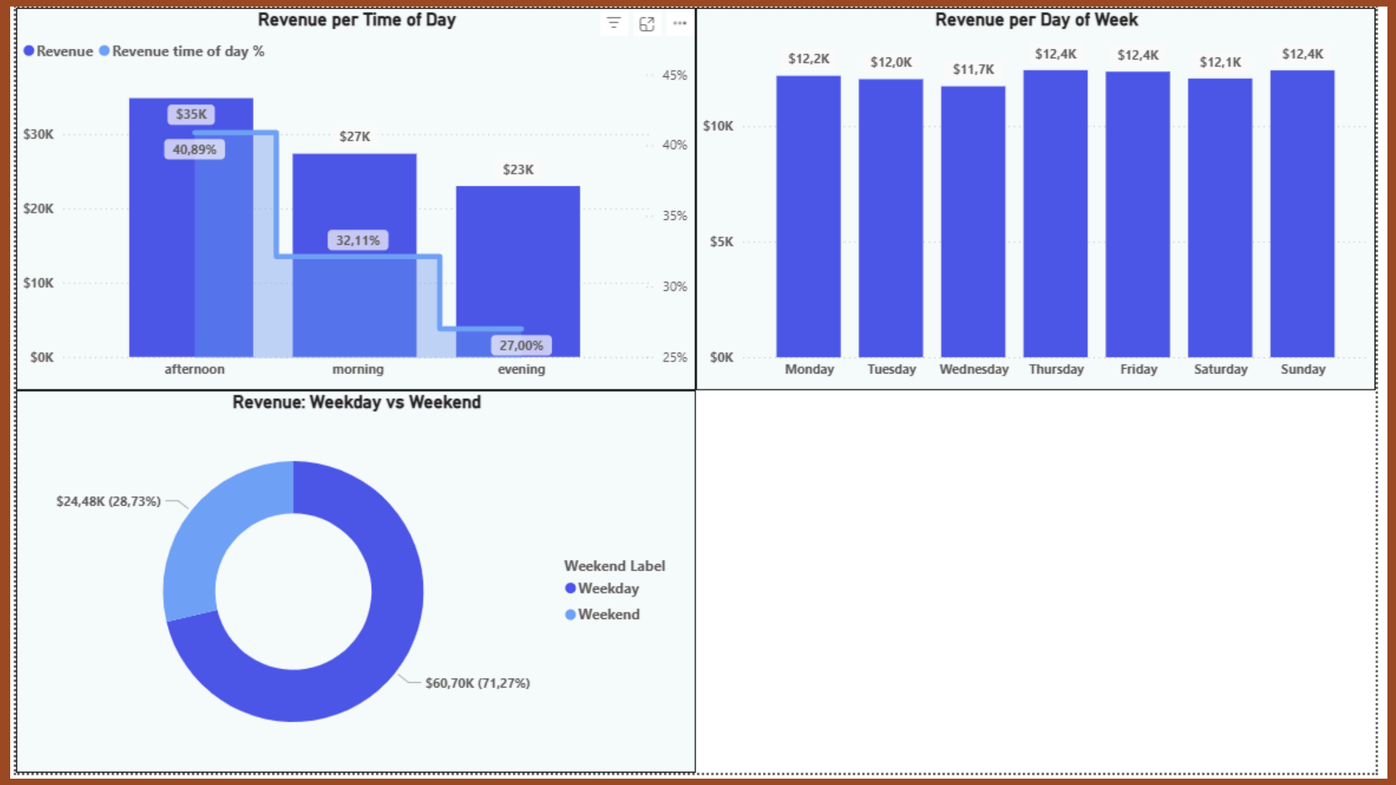




Exploratory Data Analysis

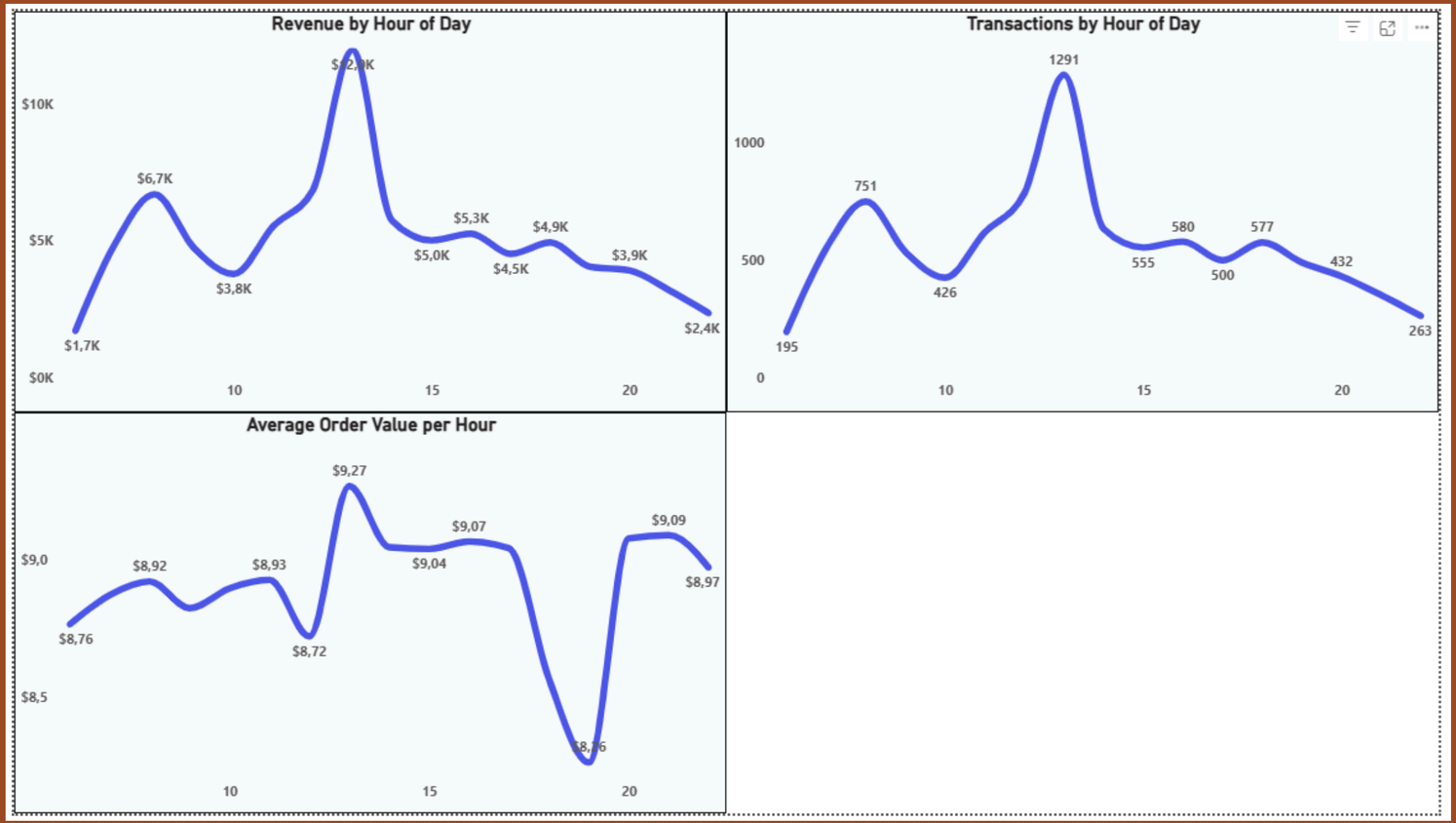
Daily revenue is quite stable, with no clear weekly or seasonal trends. Because of this, the model struggles to find strong patterns and performs close to random guessing. Adding features like monthly revenue trends could help improve its accuracy.

Our objective was to predict whether a given day would be high- or low-revenue based on temporal features, including day of the week, month, and recent revenue history.



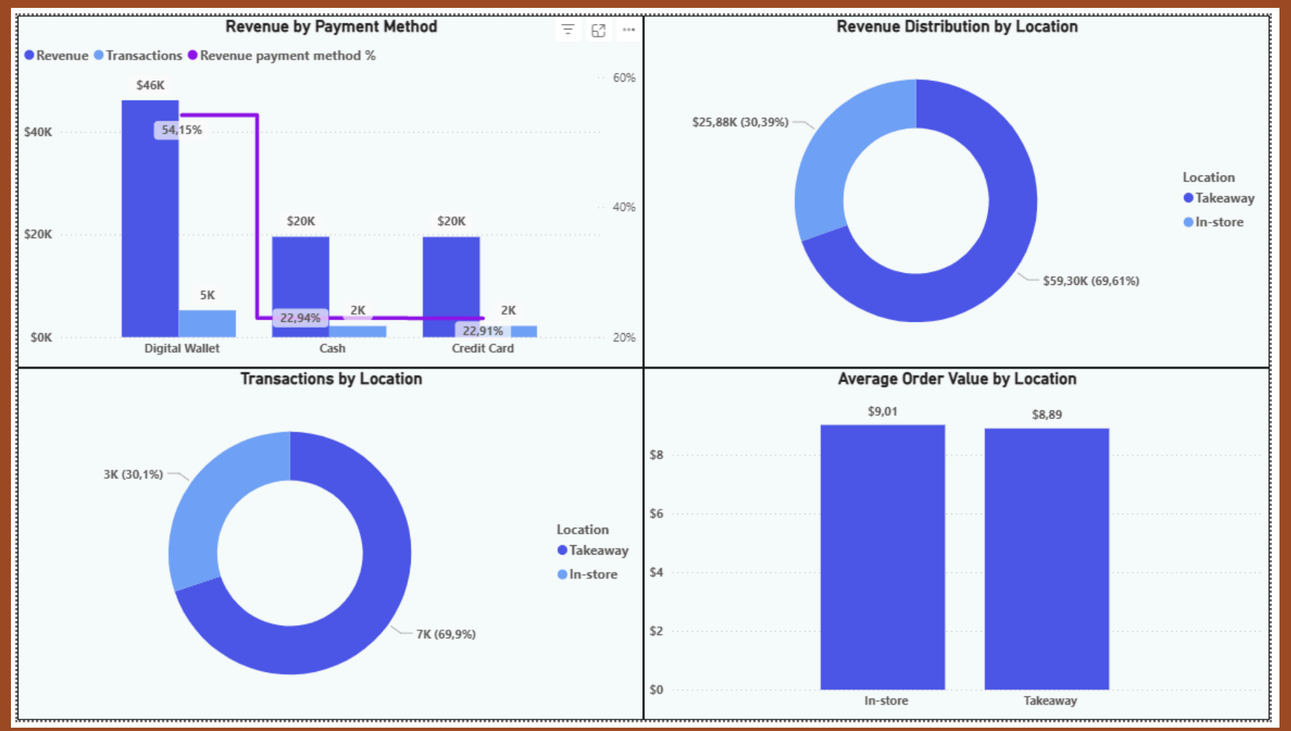
Exploratory Data Analysis

The analysis shows that most transactions and revenue occur in the afternoon, making it the busiest period of the day. Mornings are moderately active, while evenings tend to be quieter. This highlights a strong daily pattern in customer behavior.



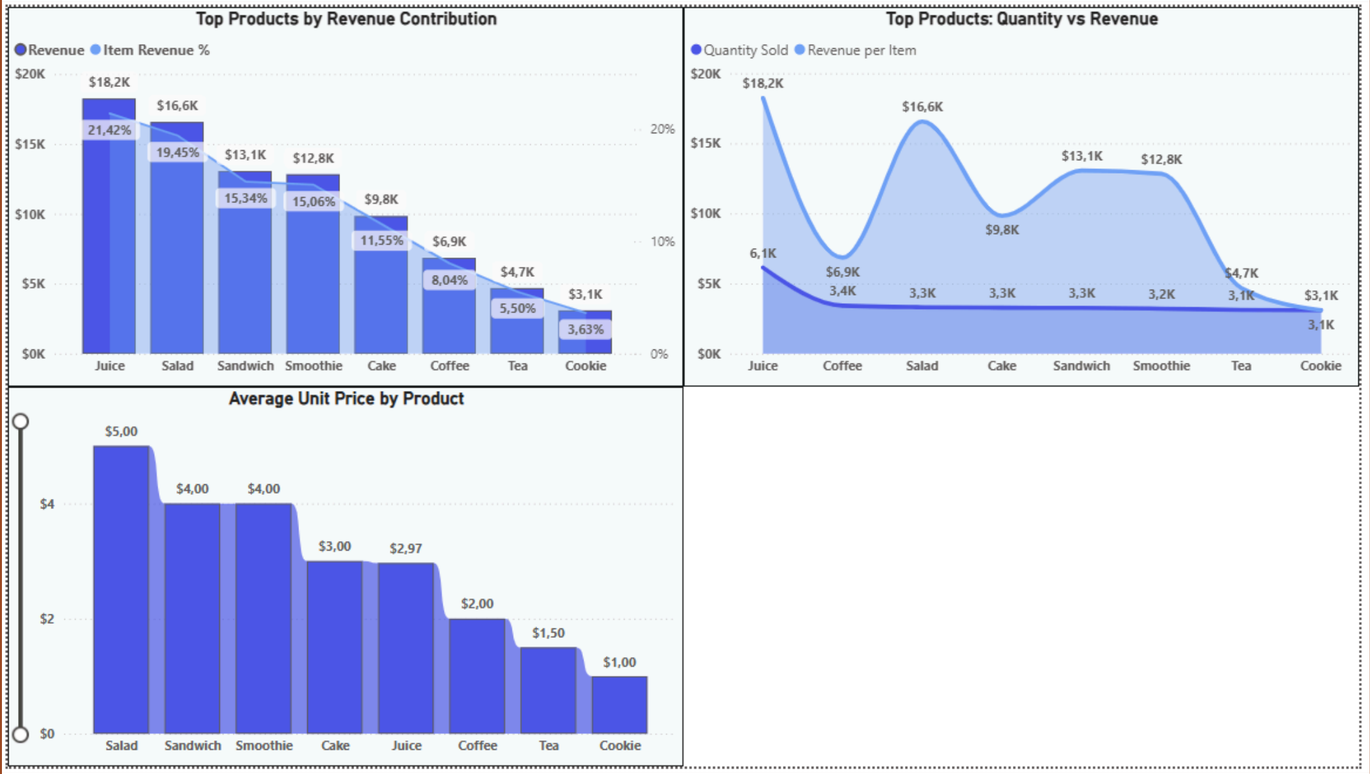
Exploratory Data Analysis

The analysis shows that takeaway orders generate significantly more revenue than in-store purchases, highlighting their importance to overall performance. Additionally, average order values vary across items, with products like salads and sandwiches contributing higher spending. This suggests that both location and product mix play a key role in driving revenue.



Modeling Approach

We developed three classification models to predict whether a given day would be busy or quiet for the café, using calendar-driven patterns such as day of the week, seasonality, and recent revenue trends.

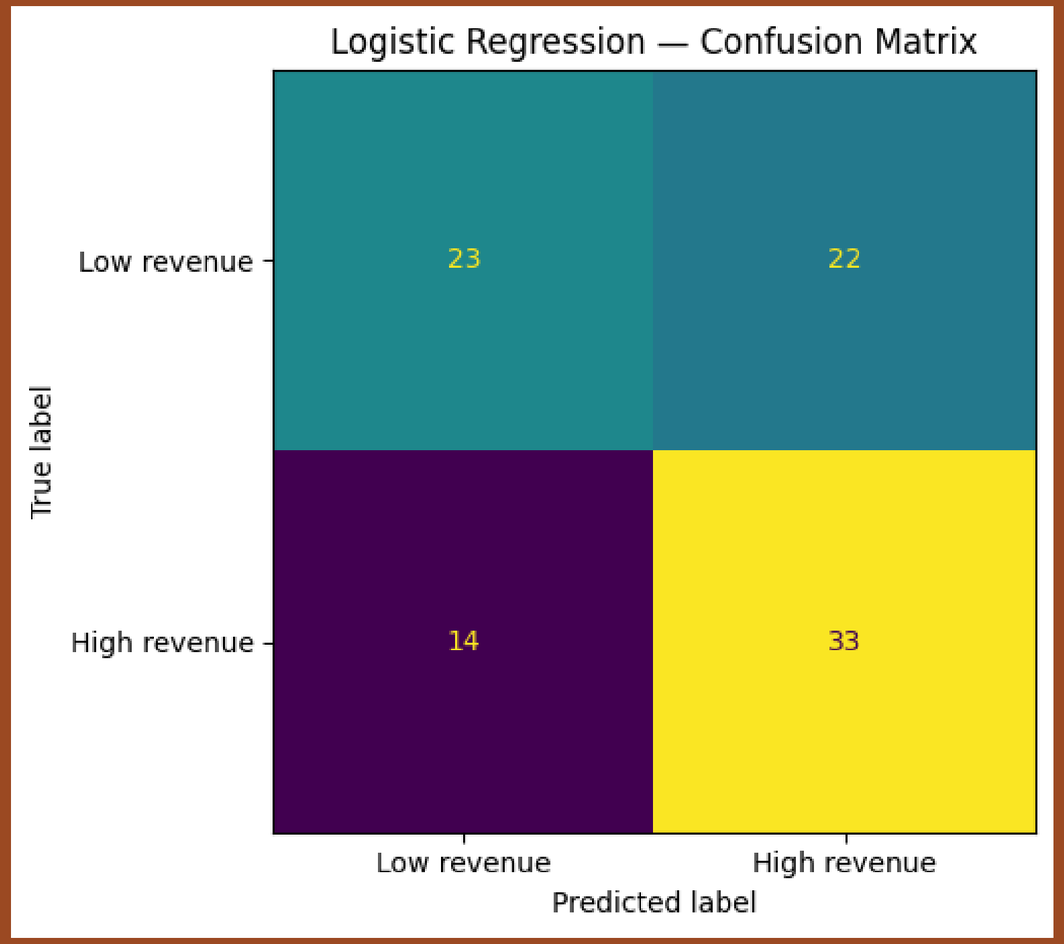


Results - Logistic Regression

Moderate performance (60.9%); biased toward predicting high-revenue days due to limited variation in revenue data.

Train/Test Split:

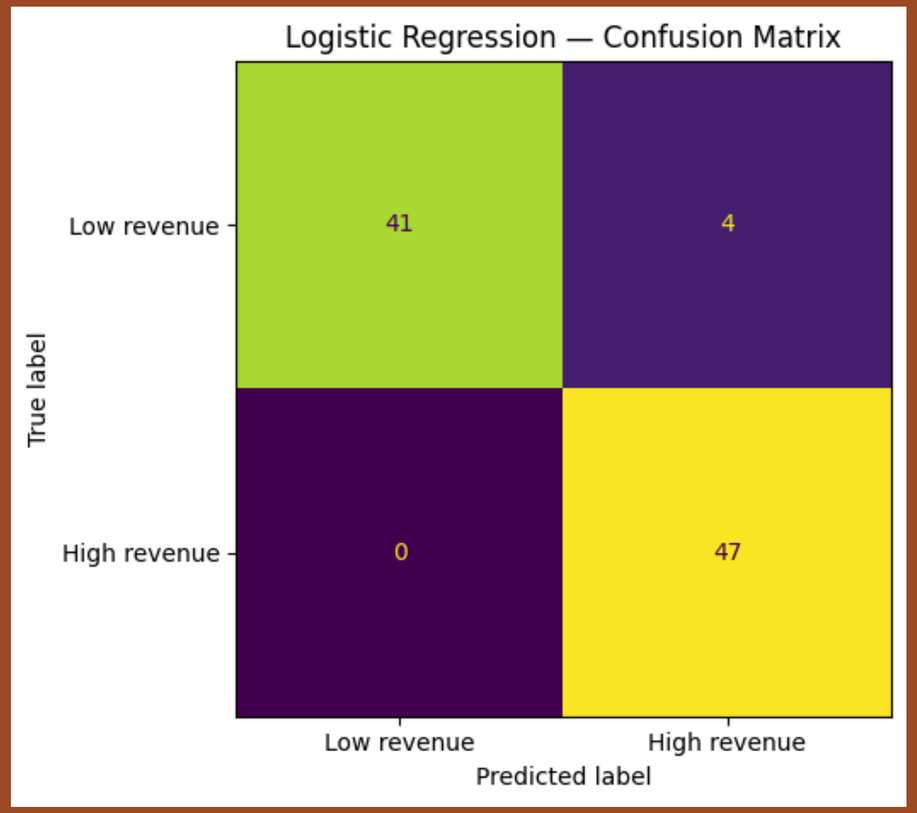
Train: Jan–Sep 2023
Test: Oct–Dec 2023
Time-based split to avoid data leakage



Confusion matrix

Results - Logistic Regression with more features

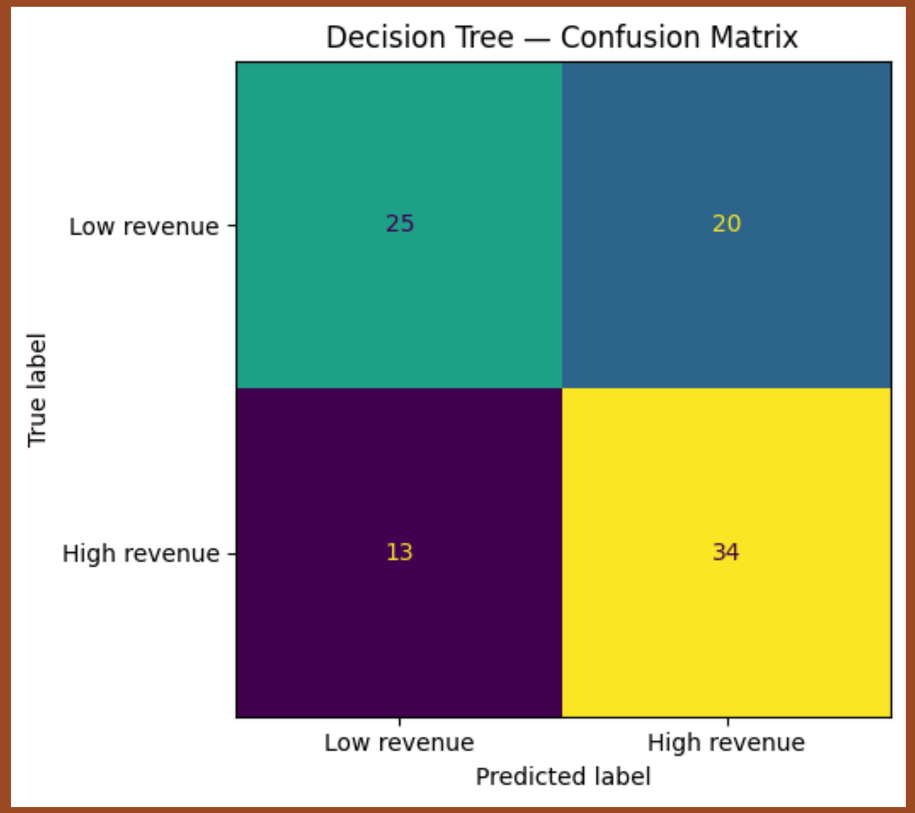
Expanding the model from 7 to 21 features resulted in an unusually high accuracy of 96%, indicating possible overfitting. This suggests the model may be memorising the data rather than generalising well. A potential solution is to include more data over multiple years to improve robustness.



Confusion matrix

Result - Decision Tree

The Decision Tree achieved moderate performance with limited overfitting. It relies primarily on recent revenue trends rather than calendar features, indicating that short-term patterns are the strongest predictors.



Confusion matrix

Modeling Approach - Random Forest

The Random Forest model is an ensemble method that combines multiple decision trees to improve robustness. It uses the same feature set as the Decision Tree model and is trained using a time-based split to preserve temporal order and avoid data leakage. This ensures the model is evaluated on future, unseen data.

Results:

- Train accuracy: 84.6%
- Test accuracy: 63.0%

Modeling Comparison

Model Performance

- Logistic Regression → ~61% accuracy
- Decision Tree → 64.1% accuracy
- Random Forest → 63.0% accuracy

The results show that all models achieve similar performance, indicating that increasing model complexity does not significantly improve accuracy. This suggests that the main limitation lies in the lack of strong predictive features rather than the choice of algorithm. Overall, the findings highlight that feature engineering plays a more important role than model selection in this task.

Findings

The café's daily revenue remains highly consistent, with most days clustering around £230. There are no clear or repeatable patterns linked to weekends, months, or seasons that distinguish stronger days from weaker ones. Because the data shows very little variation over time, the models struggled to identify meaningful relationships or predictive signals.

As a result, the best-performing model achieved an accuracy of approximately 64%, correctly predicting outcomes in about 6 out of 10 cases. While this is an improvement over random guessing, it is not sufficiently reliable to support confident business decision-making.

Business Advice

Around 70% of orders are takeaway, yet the café likely allocates most of its space and staff focus to the 30% of customers who dine in. This suggests a clear opportunity to optimise the takeaway experience—such as improving speed, pickup flow, or dedicated service areas—to better match actual demand.

The café also appears to have a loyal customer base, which creates an opportunity to drive growth by serving existing customers more effectively. This could include tailoring offerings by time of day, enhancing the takeaway channel, and beginning to collect richer customer and transaction data. Over time, this would enable more advanced analytics and stronger predictive insights.